30 August 2023

**DR. RAUL G. BRADECINA**
President
Partido State University
San Juan, Camarines Sur

**ATTENTION: Prof. Bernadette G. Gumba**
Project Leader

Dear **Dr. Padolina:**

We are pleased to inform you that the request for item realignment and change in implementation date of the project entitled, ***"Maternal and Child Health Services vis-à-vis Gender Based Belief System"*** under the Regional Research Fund initiative have been approved. The implementation of the said project now covers the period **1 September 2023** to **30 June 2024.**

In view thereof, the schedule of activities of the above-mentioned project is amended as follows:

| Target | Expected Outputs | Due Date | Amount of Release |
|---|---|---|---|
| Signing of Memorandum of Agreement (MOA) | • Signed Memorandum of Agreement | Immediately after signing of MOA | **PHP 421,317.50** |
| ***First quarter (September 1-November 30, 2023)***<br><br>Secondary data gathering and KII from person-in-charge at the MHO, RHU, and BHS of Garchitorena, Presentacion, and Siruma | Quarterly technical progress reports covering the period September 1 – November 30, 2023.<br><br>Quarterly financial report, including report of checks issued or report of cash disbursed supported with original receipts | December 31, 2023 | - |
| ***Second quarter (December 1, 2023 - February 28, 2023)***<br><br>Data cleaning and call back for all three study areas<br><br>Interview with target respondents in Garchitorena, Presentacion, and Siruma (reproductive age mother and father, and decision makers and implementers | Quarterly technical progress reports covering the period December 1, 2023 – February 28, 2024<br><br>Quarterly financial report, including report of checks issued or report of cash disbursed supported with original receipts | March 31, 2024 | **PHP 124,915.00** |

| | | | |
|---|---|---|---|
| ***Third Quarter (March 1 - June 30, 2024)***<br><br>Data cleaning and call back for all three study areas<br><br>Data processing, interpretation, and analysis<br><br>Report writing and submission | Quarterly technical progress reports covering the period March 1 – June 30, 2024.<br><br>Quarterly financial report, including report of checks issued or report of cash disbursed supported with original receipts | July 31, 2024 | - |
| Report submission | Final Technical Report and Paper in Publishable Format<br><br>Audited Financial Report | July 31, 2024<br><br>Within one month after completion of the project | **PHP 37,600.00** |
| | | **TOTAL** | **PHP 583,832.50** |

Please confirm acceptance of conditions set forth in this grant by signing on the space below and return to us the original signed conforme letter as this will form part of the official documents for the project.

Thank you and we look forward to a successful implementation of the project.

Very truly yours,

**JAIME C. MONTOYA, MS,MSc,PhD, CESO II**
*Executive Director III*

CONFORME:

**PROF. BERNADETTE G. GUMBA**
Project Leader and Dean of Graduate School
School of Graduate Studies
Partido State University

**DR. RAUL G. BRADECINA**
President
Partido State University
San Juan, Camarines Sur

**DIR. ROMMEL R. SERRANO**
Regional Director
Department of Science and Technology – Region V

DOST Form 2 (for Basic/Applied Research)

# DETAILED RESEARCH & DEVELOPMENT PROJECT PROPOSAL

**(1) PROJECT PROFILE**

Program Title:

Project Title: **MATERNAL AND CHILD HEALTH SERVICES VIS-À-VIS GENDER-BASED BELIEF SYSTEMS**

Project Leader/Sex: **Bernadette G. Gumba, PhD, Female**

Project Duration (number of months): **Ten (10)**

Project Start Date: **May 2, 2023**

Project End Date: **February 28, 2024**

Implementing Agency (Name of University-College-Institute, Department/Organization or Company): **Partido State University**

Address/Telephone/Fax/Email (Barangay, Municipality, District, Province, Region): **Goa, Camarines Sur, op@parsu.edu.ph**

**(2) COOPERATING AGENCY/IES: Local Government Unit and Municipal Health Office of Garchitorena, Presentacion, Sagnay, all of Camarines Sur**

**(3) SITE(S) OF IMPLEMENTATION**

| IMPLEMENTATION SITES NO. | COUNTRY | REGION | PROVINCE | DISTRICT | MUNICIPALITY | BARANGAY |
|---|---|---|---|---|---|---|
| 1. | Philippines | Bicol | Camarines Sur | Partido | Garchitorena | |
| 2. | Philippines | Bicol | Camarines Sur | Partido | Presentacion | |
| 3. | Philippines | Bicol | Camarines Sur | Partido | Siruma | |

**(4) TYPE OF RESEARCH**

_____ Basic

__X__ Applied

**(5) R&D PRIORITY AREA & PROGRAM (based on HNRDA 2017-2022)**

_____ Agriculture, Aquatic and Natural Resources
Commodity: _____

__X__ Health
Priority Topic: __GENDER EQUALITY___

_____ Industry, Energy and Emerging Technology
Sector: _____

_____ Disaster Risk Reduction and Climate Change Adaptation

_____ Basic Research
Sector: _____

| | |
|---|---|
| **Sustainable Development Goal (SDG) Addressed** | GOOD HEALTH & WELL-BEING, GENDER EQUALITY |

**(6) EXECUTIVE SUMMARY** (not to exceed 200 words)

The study aims to examine the relationship of the gender-based belief systems to the maternal and child health services of selected communities in the Partido District, Camarines Sur. Specifically, it aims to: (a) audit the maternal and child health programs and services of the selected study areas; (b) monitor the measurable health indicators at the barangay level; (c) test the gender-based belief systems of selected reproductive-age mothers, reproductive-age fathers, decision-makers and implementers of maternal and child care services; and (d) analyze the relationship of gender-based belief systems to the dispensation and acceptance of maternal and child health services. It will focus on gender belief systems and relate these to the outcome and outputs of maternal and child health services of the local government. It shall use a combination of written document analysis and examination of data from primary sources. One-on-one interviews will be conducted with three (3) clusters of respondents using a structured questionnaire which includes a gender-bias test. The three (3) clusters are the beneficiaries, decision-makers; and implementers of the program.

**(7) INTRODUCTION**

This research is based on the philosophy that maternal and child health are basic human rights. Healthy women are at the core of healthy societies (Marquez, 2017). The well-being of mothers, infants, and children determines the health of the next generation and can help predict future public health challenges for families, communities, and the medical care system (USDH, 2021). The health of women is not, however, innate to any society. The Philippines has the fifth highest maternal mortality in the ASEAN. Our nation is one with other countries in instituting programs and policies to protect women and children. Despite this, the effectiveness of these measures cannot be established as patriarchy remains evident in Philippine culture (Medina et al, 2015).

The proposed study shall look into maternal and child health services through a gender lens. It aims to examine the relationship of the gender-based belief systems of couples, decision-makers and implementers of health programs to the maternal and child health services of selected communities in the Partido Disctrict, Camarines Sur. It shall be a combination of written document analysis and examination of data from primary sources. One-on-one interviews will be conducted with three (3) clusters of respondents using a structured questionnaire which includes a gender-bias test culled from existing literature. The gender-bias test scores will be correlated with data on indicators of maternal and child health as enumerated above. The answers on the open-ended questions will be used to validate the gender-bias scores as well as to further determine the influence of gender-belief systems on the acceptance, response and dispensation of maternal and child health services.

**(7.1) RATIONALE/SIGNIFICANCE** (not to exceed 300 words)

The proposed study shall look into maternal and child health services through a gender lens. Studies of the OHCR (2021) maintained that deaths and injuries suffered by women and adolescent girls in pregnancy and childbirth, as well as by infants and young children, are largely preventable, yet still occur at alarming rates. The research problem at hand was prompted by the assertions of Patricio V. Marquez (2017) of the Johns Hopkins University Bloomberg School of Public Health, that social conditions and cultural norms that limit women's access to health services, education, and economic opportunities are at the root of women's health disparities and exacerbate the feminization of poverty as measured by the higher percentage of female-headed households who are poor. More of these findings are presented in the next section "Review of Literature." The proposed study aims to examine the relationship of the gender-based belief systems of couples, decision-makers and implementers of health programs to the maternal and child health services of selected communities in the Partido District, Camarines Sur.

**(7.2) SCIENTIFIC BASIS/THEORETICAL FRAMEWORK**

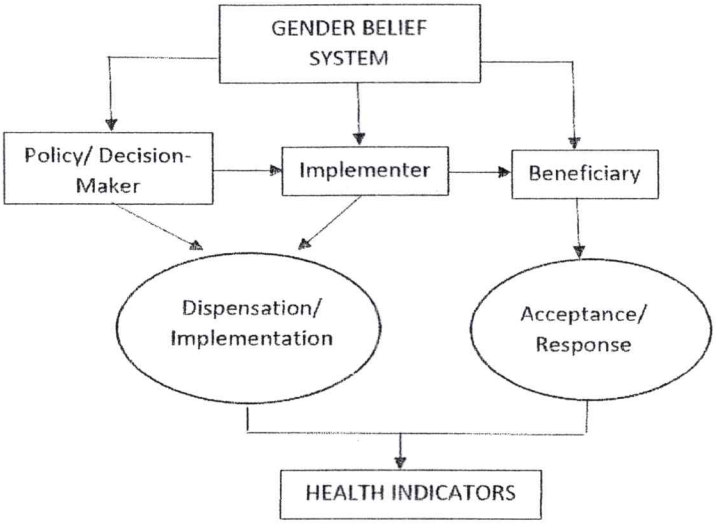



Figure 1. Maternal and Child Health Services and the Gender Belief System of Decision-Maker, Implementer and Beneficiary

A belief system is a mindset and way of thinking that manifest in a person's behavior, attitude, and way of dealing with people and structures, and formulating opinions and decisions on issues surrounding the individual (Rettig, 2017). As shown in Figure 1, this study shall examine the gender mindset of the three major stakeholders in the maternal and child health services provided by the local government. The policy and decision-maker provides the structure and guideline of the services which become the basis of implementation, the implementer dispenses the services, and the beneficiary receives the services. In all these stages, the gender belief system of the individual influences one's behavior, attitude, social interaction, and decisions – the policies that the government officials make, the way the services are executed by the implementer, and the response of the beneficiary, as well as the how the implementer and beneficiary deal with each other. All these may influence the expected outcome or the health indicators which are discussed in detail under the Objectives.

**(7.3) OBJECTIVES**

General objective: The proposed study aims to examine the relationship of the gender-based belief systems of couples, decision-makers and implementers of health programs to the maternal and child health services of selected communities in the Partido Disctrict, Camarines Sur.

Specific objectives: To accomplish the general objective, the study will

1) Audit the maternal and child health programs and services of the barangays of Garchitorena, Presentacion, and Siruma;

2) Monitor the measurable health indicators at the barangay level, as follows: no. of maternal health services, no. of child health services, no. of home visitation per month, no. of MCHS medicine dispensed per month, maternal mortality rate, infant mortality rate, premature and mature births, nutrition and malnutrition rate, and immunization rate;

3) Test the gender-based belief systems of selected reproductive-age mothers, reproductive-age fathers, and reproductive ouples;

4) Test the gender-based belief systems of decision-makers and implementers of maternal and child care services in the local communities;

5) Analyze the relationship of gender-based belief systems to the implementation and acceptance of maternal and child health services.

**(8) REVIEW OF LITERATURE**

This research is based on the philosophy that maternal and child health are basic human rights. According to the Partnership for Maternal, Newborn and Child Health (PMNCH, 2013), women's empowerment and equality is a fundamental human right and critical to achieve development objectives, including health. Healthy women are more able to actively participate in society and markets and take collective action to advance their own interests. They are likely to have greater bargaining power and control over resources within the household.

Healthy women are at the core of healthy societies (Marquez, 2017). The well-being of mothers, infants, and children determines the health of the next generation and can help predict future public health challenges for families, communities, and the medical care system (USDH, 2021).

Findings from new research provide evidence about the impact that the well-being of women has on the intergenerational propagation of good physical and mental health (Marquez, 2017). Women, who are key in maintaining healthy families, access the health system more than men, both for themselves and on behalf of their children. Many become pregnant and give birth, a significant health event, then typically become their child's primary caregiver, a role that greatly influences household health overall (Wheeler et al, 2013). The findings published in 2016 indicate that pregnancy is associated with substantial changes in brain areas of pregnant women that are responsible for social cognition and the ability to understand the thoughts and intentions of others and that they may help intensify the bonding between mothers and their babies, and hence, the survival and healthy development of children (Marquez, 2017). Moreover, healthy birth outcomes and early identification and treatment of health conditions among infants can prevent death or

disability and enable children to reach their full potential (USDH, 2021). Corollarily, a healthy woman is equal to a healthy family and this ultimately leads to a healthy society.

The health of women is not, however, innate to any society. Development experience has shown that deliberate policies and programmatic strategies aimed at nurturing women's health and well-being across the life cycle are vital for realizing the full potential of women and girls (Marquez, 2017).

The World Bank Group Gender Strategy (WBG, 2015) pointed that while some women may face fewer disadvantages now than in the past, major gaps remain. One significant gap is excess female mortality, which could be avoided with better access to quality health care services, particularly during pregnancy and child deliver. Social conditions and cultural norms that limit women's access to health services, education, and economic opportunities are at the root of women's health disparities and exacerbate the feminization of poverty as measured by the higher percentage of female-headed households who are poor. Research has shown that many maternal illnesses and lifestyle behaviors also affect children, amplifying their negative impact on society (Marquez, 2017).

It should be clear that addressing the root causes of ill health, premature mortality, and disability among women, and enhancing women's voice and agency to deal with limiting, and in some cases, oppressive, social and cultural norms and gender-based violence that impede the health and human capacity development of women is a cross-sectoral development priority. Policies that help turn the tide against the feminization of poverty and toward enabling women to lead lives of sustainable economic advancement and self-reliance will at the same time improve the health and mental well-being of future generations, and contribute to ensure that development is socially and economically inclusive (Marquez, 2017).

Based on statistics, maternal mortality rate in the Philippines has been steadily declining. In 2017, maternal deaths were 121 per 1,000 live births or a 2.42% decline from 2016. In 2016, it was 124 or a 2.36% decline from 2015. In 2015, it was 127 or a 3.05% decline from 2014 (Evelino & Benegas, 2015). However, the Philippines has the fifth highest maternal mortality in the ASEAN, with Lao PDR being the first with 220 maternal deaths per 1,000 live births (ASEAN, 2015). The Philippines, as a signatory of various international declarations, is one with other countries in instituting programs and policies to protect women and children. Despite this, the effectiveness of these measures cannot be established as patriarchy remains evident in Philippine culture (Medina et al, 2015).

Below is a table of the magnitude of maternal mortality in selected areas of Camarines Sur, derived from a recent study of De Vergara et al (2019).

Table 2. Magnitude of maternal mortality in Camarines Sur, 2013-2018 (De Vergara et al, 2019)

| Area Category | 2013 | 2014 | 2015 | 2016 | 2017 | 2018 |
|---|---|---|---|---|---|---|
| Rural | 763 | 1471 | 1770 | 1140 | 2400 | 4717 |
| Coastals | -- | -- | -- | -- | 999 | 460 |
| Island | -- | -- | -- | -- | -- | 1000 |
| Upland | -- | -- | -- | -- | 999 | 0 |

The study of De Vergara et al (2019) covered four (4) municipalities of Camarines Sur, namely: Garchitorena, Presentacion, San Jose and Calabanga. Its findings included the magnitude of maternal mortality in 2013-2018. The data shows an increasing maternal mortality particularly from 2013 to 2018 with 2,400 and 4,717 in the latter years of 2017 and 2018. There was a data deficient period in 2013-2015 while information was more available in 2016-2018. So the table above shows a relatively reliable set of information in the latter period.

## (9) METHODOLOGY

*Study design – this section indicates how the study objectives will be achieved. It includes a description of the type of study design eg. Cross sectional, case control, cohort, etc.*

The study will use the qualitative method as it will focus on gender belief systems and relate these to the outcome and outputs of maternal and child health services of the local government. It shall use a combination of written document analysis and examination of data from primary sources. One-on-one interviews will be conducted with three (3) clusters of respondents using a structured questionnaire which includes a gender-bias test culled from existing literature. The three (3) clusters are as follows: (a) the beneficiaries of the maternal and child health services; (b) the decision-makers; and (c) the implementers of the program.

The gender-bias test scores will be analyzed in reference to the data on indicators of maternal and child health as enumerated above. The answers on the open-ended questions will be used to validate the gender-bias scores as well as to further determine the influence of gender-belief systems on the acceptance, response and dispensation of maternal and child health services.

*Study population – this is required for studies involving animals and humans. This section states the number of study subjects required to enter and complete the study. A brief definition of the type of study subject required is also described.*

The respondents will be selected from three municipalities, namely Siruma with 22 barangays, Garchitorena with 23 and Presentacion with 18. The respondents will be clustered as beneficiaries, decision-makers and implementers. For the beneficiaries, there will be 100 respondents. For the decision-makers, there will be 30, preferably officials who are directly involved in health committees. For the implementers, 30 rural health personnel or barangay health workers will be selected.

Respondents will be selected at random using multi-stage sampling method and the random tool of the Microsoft Excel Program.

*Inclusion criteria – this section describes the criteria each study subject must satisfy to enter the study. These criteria may include, but are not limited to the following: age, sex, race, diagnosis/condition, method of diagnosis, diagnostic test.*

Reproductive-age mothers are individuals aged 18-45 who have experienced pregnancy, whether full or partial, whether live birth or otherwise, whether normal delivery or otherwise, etc. Reproductive-age fathers are husband or partner of a reproductive-age mother. The respondents may be any of the following: with children, with a child who died at birth, with premature birth, with mature birth, with a wife who had difficult pregnancy, with a wife who had easy pregnancy, and/or a man who lost his wife to pregnancy. The respondents are living in the barangay covered by the study, able to speak and understand the local language, and mentally fit.

Decision-makers are local government officials who directly participate in making decisions pertinent to the delivery of maternal and child health services in their locality. Implementers are personnel of Rural Health Units or barangay health centers who directly dispense maternal and child health services.

For Questionnaire 1 which will be utilized to gather secondary data, the personnel-in-charge do not need to answer every question as long as the folders and reports will be provided to the enumerator. The clarificatory questions will be listed down by the enumerator which shall give the respondent around 20-30 minutes to answer. For Questionnaire 2, the respondents will be able to answer fully in 30-40 minutes each.

*Exclusion criteria – this section details the criteria that would eliminate a study subject from participation in the study.*

Excluded in the study are those who have current symptoms of physical weakness or illness, or life-threatening co-morbidities, or signs of dementia or psychosis.

*Withdrawal criteria – this section details the criteria that would make a study subject unfit to participate at the time of the conduct of the study, as well as the indicators in which a study subject can withdraw from participating.*

Respondents who were initially identified shall be allowed to withdraw upon request, as well as study

subjects who suddenly got sick or who have to brought to the doctor, or who have to be somewhere else at the time of the study.

*Sample size computation – this section describes the type of sampling design and the assumptions used to compute the sample size.*

The table below shows the total number of households per municipality covered, the number of sample barangays for each, the number of sample beneficiaries, decision-makers and implementers. The household data were taken from the website of the Philippine Statistics Authority (PSA, 2016). The distribution of samples was based on the percentage share of each municipality on the total number of households i.e. 40% for Garchitorena, 31% for Presentacion and 29% for Siruma.

Table 3. List of Municipalities, number of households and sample respondents per cluster

| Municipality | Total no. of households | Percent | No. of sample beneficiaries | No. of sample decision-makers | No. of sample implementers |
|---|---|---|---|---|---|
| Garchitorena | 5216 | 40 | 40 | 12 | 12 |
| Presentacion | 4048 | 31 | 31 | 9 | 9 |
| Siruma | 3789 | 29 | 29 | 9 | 9 |
| | 13044 | 100 | 100 | 30 | 30 |

The sample size of the beneficiaries was determined based on the Criteria Sampling Technique (Shaheen et al., 2018). This sampling method is used to collect data from purposeful samples which, in the context of this study, will be the reproductive-age mothers and fathers as defined earlier. This specific criterion would exclude a relatively substantial portion of the population such as those outside the reproductive age, or those whose children are already beyond 5 years old, or those who have no children. According to Shaheen et al. (2018), this sampling method is strategic because it would yield the best information and have an impact on the findings. Logical generalizations can be made from the evidence provided by the sample.

The sample size of the decision-makers and implementers was mush fewer, which was determined based on the Typical Case Sampling (Shaheen et al., 2018). The typical cases are the knowledgeable participants. In the context of this study, the matters faced by the decision-makers on health services and the job description of health personnel are generally similar, and therefore these individuals know what is typical insofar as their functions are concerned.

For the selection of the beneficiary-respondents, a multi-stage sampling will be used specifically the random starting point, right-shoulder rule and Kish grid. For the selection of the decision-makers and implementers the random tool of the Microsoft Excel Program will be utilized.

***Site of the study – this section details the location, station or unit where the R and D will be conducted***

The study shall cover three (3) of the poorest municipalities in Camarines Sur. From Garchitorena, two (2) island barangays and two (2) poblacion barangays will be chosen. From Siruma, three (3) coastal barangays will be selected. From Presentacion, three (3) upland barangays will be picked.

Based on the "City and Municipal-Level Small Area Poverty Estimates of 2015" issued by the Philippine Statistics Authority (PSA), the municipalities of Garchitorena, Siruma and Presentacion were among the top three poorest in the province. Below is an excerpt of the PSA data:

Table 4. Top 10 poorest municipalities in the province of Camarines Sur

| Rank | Muicipality | Poverty incidence | Rank | Municipality | Poverty incidence |
|---|---|---|---|---|---|
| 1 | Garchitorena | 59.7 | 6 | Caramoan | 51.3 |
| 2 | Siruma | 59.2 | 7 | Bula | 45.4 |
| 3 | Presentacion | 52.8 | 7 | Minalabac | 45.4 |
| 4 | Balatan | 51.7 | 8 | Del Gallego | 44.7 |
| 5 | Tinambac | 51.4 | 9 | Ragay | 44.6 |

A high poverty incidence leads to many other dismal circumstances. Households cannot afford their food and non-food needs, including health and sanitation. If consumers are poor, businesses are not encouraged to invest in the locality. With poor residents and without new businesses, the local government will collect very minimal amount of taxes. With a small tax revenue, the government will lack the funds to finance its programs, including maternal and child care. Moreover, the government will face great difficulty in setting its priorities. It may opt to give preference to food and employment problems first before maternal and child health.

The proponent and research team will be based at Partido State University in Goa, Camarines Sur under the auspices of the BCHRD.

***Study plan – this section explains the plan of action, procedures and methods to be used during the study. Detailed methodology is described for laboratory, diagnostic, interviews, manner of data collection. Special instrumentation may be described in a subsection (instrumentation/data collection tools, special equipment, etc.)***

The audit of existing maternal and child health services and measurable health indicators will be gathered from secondary sources and validated through key infomratn interview.

The gender-based belief systems will be surveyed in three clusters: the beneficiaries of the maternal and child health services, the decision-makers and the implementers of the program. The beneficiaries will be selected from reproductive-age mothers, reproductive-age fathers and reproductive couples. The decision-makers will be from among the barangay officials and the program implementers will be from the personnel of the Rural Health Unit or Barangay Health Station.

To test the gender-based belief systems a one-on-one interview will be conducted using a standard tool. The questionnaire will have two parts, namely: (a) a gender-bias test; and (b) open-ended questions to determine if the respondent's point of view influences the response, acceptance or dispensation of maternal and child health care services.

The gender-bias test is a localized and consolidated version of the Harvard University's Implicit Gender-Career Bias Test and 2015 Commonwealth of Learning Gender-Bias Quiz. The tool will be revised to fit the context of rural and relatively low-income living, and translated to Bicol language. Stratified sampling will be used so that there will be from reproductive couples, MHO personnel, RHU personnel, and LGU officials in the health committees.

Pearson's Correlation Analysis will be utilized to find out if the compounded test scores are associated with the indicators of maternal and child health program as enumerated above.

***Case report form – the case report form (CRF) should be attached to the research proposal. If the CRF is in electronic format, a printed copy should be attached as an appendix.***

***Variables to be investigated – dependent/outcome and independent variables***

Pearson's Correlation Analysis will be used to test the relationship of two (2) sets of variables. The first, as enumerated below, shall be the measurable outcomes and outputs of the health services.

- No. of maternal health services
- No. of child health services
- No. of home visitation per month
- No. of MCHS medicine dispensed per month
- Maternal mortality rate
- Infant mortality rate
- Number of premature births
- Number of mature births
- Nutrition rate
- Malnutrition rate
- Immunization rate;

The second set will be the results of the gender-bias quiz that will be administered through one-on-one interview with the target respondents, as discussed in previous sections.

*Plans for data processing and analysis*
- *Computer facilities to be used, software packages*
- *Statistical tools/tests to be used*
- *Dummy tables*

The data will be processed using frequency, percentage, scoring of gender-bias quiz, list of gender-based belief systems (gender sensitive vs. gender biased), Pearson's Correlation Analysis between (a) maternal and child health indicators; and (b) gender-bias scores.

**(10) TECHNOLOGY ROADMAP** (if applicable) (use the attached sheet)

NOT APPLICABLE

**(11) EXPECTED OUTPUTS (6Ps)**

The project shall produce the following expected outputs based on the 6Ps:

Publication: The research paper will be converted into a publishable form, then presented in an academic conference and published in a refereed journal
Patent: Not Applicable
Product: Not Applicable
People Service: Concrete recommendation to policy-makers and implementers for a more effective and gender-sensitive implementation of maternal and child health services; IEC Materials and gender sensitivity training for MHO, RHU and BHW personnel
Place and Partnership: Summarized research report for presentation to concerned LGUs; Extension project proposal for future partnership among PSU, LGU and MHO on maternal and child health programs
Policy: Policy recommendation to LGU and MHO, and additional guidelines in the delivery of maternal and child health services

Furthermore, the proposed study shall provide the following outputs:

(a) Report on maternal and child health program of Garchitorena, Siruma and Presentacion;
- No. of maternal health services
- No. of child health services
- No. of home visitation per month
- No. of MCHS medicine dispensed per month
- Maternal mortality rate
- Infant mortality rate
- Number of premature births
- Number of mature births
- Nutrition rate

- Malnutrition rate
- Immunization rate;

(b) Report on the results of the gender-bias test results of selected reproductive-age mothers and fathers, reproductive couples, selected local government officials, and RHU personnel or barangay health workers.

(c) Results of the statistical analysis of maternal and child health program indicators and the gender-bias test scores

(d) Policy proposal

- Analysis of the relationship of gender-based belief systems to the implementation of child and maternal health services
- Analysis of the relationship of gender-based belief systems to the acceptance and response of beneficiaries to child and maternal health service
- Policy Recommendations

**(12) POTENTIAL OUTCOMES**

First of all, in the study areas, there shall be an estimated 30-50% decrease in maternal, infant and child mortality rates, approximately 30-50% decrease in the number of premature births, and about 30-50% decrease in malnutrition rate. Conversely, there shall be a corresponding increase in mature births, nutrition rate and immunization rate. Additionally, there shall be an increase in the number of women, infants and children who are provided health services, an increase in the frequency of home visitation, and an increase in the number of medicine dispensed for women, infants and children.

The LGU, MHO, RHU and BHW, would have established new policies, procedures, and possible changes in the delivery of maternal and child health services that would promote gender sensitivity and gender equality between couples, their family and community. The delivery of maternal and child services would actually be responsive to the gender-based belief systems of beneficiaries and will eventually lead to a healthy and strong citizenry in the long-run. The study should have served as a concrete step so that science may effectively serve the poor and the marginalized.

**(13) POTENTIAL IMPACTS (2Is)**

Social Impact: Gender-sensitive and effective delivery of maternal and child health services; empowerment of women as partners in nation-building; promote gender sensitivity and gender equality between couples and between service providers and beneficiaries; a wider perspective on the dynamics between couples and within their households so that acceptance and response to health care services may be more effective; an alternative perspective for government personnel so that they may be more sensitive and in touch with the program beneficiaries

Economic Impact: Empowerment of women as partners in economic development; strengthening of human capital through effective health services; the results of this research may put in plain view the gender-based beliefs and dynamics between couples, their households and community and with the program implementers so that the delivery of maternal and child services are actually responsive to the needs of the beneficiaries and will eventually lead to a healthy and productive citizenry in the long-run.

**(14) TARGET BENEFICIARIES**

*This refers to the probable end-users or beneficiaries of the research output and the number and locality of beneficiaries, if applicable.*

The direct beneficiaries will be reproductive-age individuals and local government personnel of the study areas. The direct benefit for all respondents will be the opportunity to assess from their own point of view the health services rendered by the government. Answering the questions in the study will help them look back, reflect and process how their participation - whether as beneficiary, decision-maker or implementer - has somehow accomplished the objectives of the maternal and child health services under consideration.

The LGU and MHO will likewise benefit because of the positive effects of improving the services that will promote the welfare of the community members especially the prospective mothers and future children. A healthy mother and child will lead to a healthy family and eventually to a healthy community. The decision-makers and implementers will gain additional information which they may opt to utilize in the continuance of health services delivery.

Moreover, for the LGU, MHO, RHU and BHS, the results of this study may provide sound basis for possible changes in the delivery of maternal and child health services, i.e. programs and activities that will promote gender sensitivity and gender equality between couples and between service providers and beneficiaries. For the implementers of child and maternal health services, this study may serve as alternative perspective so that they may be more sensitive and in touch with the program beneficiaries. For the local government units, MHO and RHU, the results of this research may put in plain view the gender-based beliefs and dynamics between couples, their households and community and with the program implementers so that the delivery of maternal and child services are actually responsive to the needs of the beneficiaries and will eventually lead to a healthy and strong citizenry in the long-run.

**(15) SUSTAINABILITY PLAN** (if applicable)

Below is a list of ways on how the project will be implemented/ operated amidst pre-identified risks. Details are presented in a separate file **DOST Form 5 C. Risk Management Plan.**

- Lack of data: Find alternative sources such as the Barangay Health Workers and the personnel of the Barangay Health Center, triangulate the data from the MHO, RHU and BHS; inquire from the provincial and regional officers of the DOH;
- Weather disturbances: Monitor weather reports and plan the fieldwork ahead to capitalize on good weather conditions;
- Unstable peace and order situation: Coordinate with the LGU and police to ensure that the communities are safe for the surveys and interviews;
- Data gaps: Conduct data cleaning and review before accepting the results of the data-gathering;
- Target respondents might not cooperate: Explain clearly the provisions in the Informed Consent Form; seek help from the barangay officials to identify other qualified respondents

**(17) LIMITATIONS OF THE PROJECT**

The project shall only cover three selected municipalities of the Partido District, Camarines Sur classified as rural and among the poorest in the province. Another perspective may have been gathered if the study would include urban and relatively well-off communities.

**(18) LIST OF RISKS AND ASSUMPTIONS RISK MANAGEMENT PLAN** (List possible risks and assumptions in attaining target outputs or objectives.)

Below is a list of possible risks and assumptions in attaining the target outputs and objectives. The risk management plan is presented in a separate file **DOST Form 5 C. Risk Management Plan.**

- Lack of data;
- Weather disturbances;
- Unstable peace and order situation
- Data gaps;
- Unstable peace and order situation;
- Data gaps;
- The enumerators are residents of the study area;
- The target respondents might not cooperate;
- Too much work at the office;
- Data collection may be facilitated by the complete documentation of health services at the different levels - provincial, municipal and barangay
- The probability of success and timely completion of the project may be ensured because the Project Leader has engaged in various research, since 2009, which were implemented based on objectives and completed on time;

**(19) LITERATURE CITED**

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**(20) PERSONNEL REQUIREMENT**

| Position | Percent Time Devoted to the Project | Responsibilities |
|---|---|---|
| **Project Leader** | **50%** | **Overall in charge of project implementation, analyze data, prepare research report** |
| **Clerk II –Project Staff** | **100%** | **Facilitate communication and coordination with government units, agencies and other stakeholders in the project, organize training for enumerator/s, organize, oversee and** |

| | | |
|---|---|---|
| | | supervise fieldwork, evaluate completeness of data, process data, report to the Project Leader |
| Cleek I - Enumerator | 100% | Coordinate with the target respondents, gather all the data required in the project, report to the Project Staff |

**(21) BUDGET BY IMPLEMENTING AGENCY**

| IMPLEMENTING AGENCY | PS | MOOE | EO | Total |
|---|---|---|---|---|
| Year 1 | 0 | P125,000.00 | 0 | P125,000.00 |
| | | | | |
| | | | | |
| TOTAL | | | | |

**(22) OTHER ONGOING PROJECTS BEING HANDLED BY THE PROJECT LEADER:** __0__ (number)

| Title of the Project | Funding Agency | Involvement in the Project |
|---|---|---|
| N/A | | |
| | | |
| | | |

**(23) OTHER SUPPORTING DOCUMENTS** (Please refer to page 2 for the additional necessary documents.)

**I hereby certify the truth of the foregoing and have no pending financial and/or technical obligations from the DOST and its attached Agencies. I further certify that the programs/projects being handled is within the prescribed number as stipulated in the DOST-GIA Guidelines. Any willful omission/false statement shall be a basis of disapproval and cancellation of the project.**

| | SUBMITTED BY (Project Leader) | ENDORSED BY (Head of the Agency) |
|---|---|---|
| Signature | | |
| Printed Name | BERNADETTE G. GUMBA, PHD | RAUL G. BRADECINA, PHD |
| Designation/Title | Professor V | SUC President III |
| Date | April 14, 2023 | April 14, 2023 |

*Note: See guidelines/definitions at the back.*

**DOST Form 2 (for Basic/Applied Research)**
**DETAILED R & D PROJECT PROPOSAL**

**I. General Instruction:** Submit through the DOST Project Management Information System (DPMIS), http://dpmis.dost.gov.ph, the detailed R&D proposal for the component project together with the detailed proposal of the whole Program, project workplan, line-item budget (LIB), 1-page curriculum vitae of the Project Leader, and Certificate of Incorporation or DTI Registration (if applicable) and other applicable supporting documents required under item II.23 below. Also, submit four (4) copies of the proposal together with its supporting documents. Use Arial font, 11 font size.

**II. Operational Definition of Terms:**

**1. Title**- the identification of the Program and the component projects.

**Project**- refers to the basic unit in the investigation of specific S&T problem/s with predetermined objective/s to be accomplished within a specific time frame.

**Project Leader**- refers to a project's principal researcher/implementer.

**Project Duration**- refers to the grant period or timeframe that covers the approved start and completion dates of the project, and the number of months the project will be implemented.

**Implementing Agency**- the primary organization involved in the execution of a program/project which can be a public or private entity

**2. Cooperating Agency/ies**- refers to the agency/ies that support/s the project by participating in its implementation as collaborator, co-grantor, committed adopter of resulting technology, or potential investor in technology development or through other similar means.

**3. Site/s of Implementation**- location/s where the project will be conducted. Indicate the barangay, municipality, district, province, region, and country.

**4. Type of Research**- indicates whether the project is basic or applied.
**Basic research**- is an experimental or theoretical work undertaken primarily to acquire new knowledge of the underlying foundations of phenomena and observable facts, without any particular or specific application or use in view.
**Applied research**- is an investigation undertaken in order to utilize data/information gathered from fundamental/basic researches or to acquire new knowledge directed primarily towards a specific practical aim or objective with direct benefit to society.

**5. R&D Priority Area and Program**- based on the Harmonized National R&D Agenda 2017-2022, indicates which R&D agenda the project can be categorized in: Agriculture, Aquaculture and Natural Resources; Health; Industry, Energy, and Emerging Technology; Disaster Risk Reduction and Climate Change Adaptation; and Basic Research. Indicate also the specific Commodity/Sector, whether crops, livestock, forestry, agricultural resources or socio-economics; fisheries or aquatic resources; biotechnical, pharmaceutical, or health services; biotechnology, information technology, material science, photonics or space technology; industry, energy, utilities or infrastructure.

**Sustainable Development Goal (SDG) Addressed**- indicates which among the 17 SDGs adopted by the United Nations Members States are addressed by the project

**6. Executive Summary**- briefly discusses what the whole proposal is about

**7. Introduction**- a formally written declaration of the project and its idea and context to explain the goals and objectives to be reached and other relevant information that explains the need for the project and aims to describe the amount of work planned for implementation; refers to a simple explanation or depiction of the project that can be used as communication material.

**7.1. Rationale**- brief analysis of the problems identified related to the project
**Significance**- refers to the alignment to national S&T priorities, strategic relevance to national development and sensitivity to Philippine political context, culture, tradition and gender and development.

**7.2. Scientific Basis**- other scientific findings, conclusions or assumptions used as justification for the research
**Theoretical Framework**- the structure that summarizes concepts and theories that serve as basis for the data analysis and interpretation of the research data.

**7.3. Objectives**- statements of the general and specific purposes to address the problem areas of the project.

**8. Review of Literature**- refers to the following: (a) related researches that have been conducted, state-of-the-art or current technologies from which the project will take off; (b) scientific/technical merit; (c) results of related research conducted by the same Project Leader, if any; (d) Prior Art Search, and; (e) other relevant materials.

**9. Methodology**- discusses the following: (a) variables or parameters to be measured and evaluated or analyzed; (b) treatments to be used and their layout; (c) experimental procedures and design; (d) statistical analysis; (e) evaluation method and observations to be made, strategies for implementation (Conceptual/Analytical framework).

**10. Technology Roadmap** (if applicable)- a visual document that communicates the plan for technology. It is a flexible planning technique to support strategic and long-range planning by matching short- and long-term goals to specific technology solutions.

**11. Expected Outputs (6Ps)**- deliverables of the project based on the 6Ps metrics (Publication, Patent/Intellectual Property, Product, People Service, Place and Partnership, and Policy).

*Publication*- published aspect of the research, or the whole of it, in a scientific journal or conference proceeding for peer review, or in a popular form.
*Patent/Intellectual Property*- proprietary invention or scientific process for potential future profit.
*Product*- invention with a potential for commercialization.
*People Service*- people or groups of people, who receive technical knowledge and training.
*Place and Partnership*- linkage forged because of the study.
*Policy*- science-based policy crafted and adopted by the government or academe as a result of the study.

**12. Potential Outcomes**- refer to the result that the proponent hopes to deliver three (3) years after the successful completion of the project.

**13. Potential Impacts**

*Social Impact*- refers to the effect or influence of the project to the reinforcement of social ties and building of local communities.
*Economic Impact*- refers to the effect or influence of the project to the commercialization of its products and services, improvement of the competitiveness of the private sector, and local, regional, and national economic development.

**14. Target Beneficiaries**- refers to groups/persons who will be positively affected by the conduct of the project.

**15. Sustainability plan**- refers to the continuity of the project or how it shall be operated amidst financial, social, and environmental risks.

**16. Gender and Development (GAD) Score**- refers to the result of accomplishing GAD checklists (for project monitoring and evaluation/project management and implementation) to highlight the contribution of the project in the achievement of the objectives of Republic Act 7192, "Women in Development and Nation Building Act," interpreted as gender-responsive, gender-sensitive, has promising GAD concepts, or GAD is invisible.

**17. Limitations of the Project**- refer to restrictions or constraints in the conduct of the project.

**18. Risk**- refers to an uncertain event or condition that its occurrence has a negative effect on the project.

**Assumption**- refers to an event or circumstance that its occurrence will lead to the success of the project.

**19. Literature Cited**- an alphabetical list of reference materials (books, journals and others) reviewed. Use standard system for citation.

**20. Personnel Requirement**- details on the position of personnel to be involved in the project, percent time devoted to the project, and responsibilities.

**21. Budget By Implementing Agency**- personnel services (PS), maintenance and other operating expenses (MOOE), and equipment outlay (EO) requirement of the project by implementing agency for

Year 1 and for the whole duration of the project. Please refer to the DOST-GIA Guidelines for the details (Section IX.B of DOST Administrative Order (A.O.) 011, s. 2020).

a. **PS**- total requirement for wages, salaries, honoraria, additional hire and other personnel benefits.

b. **MOOE**- total requirement for supplies and materials, travel expenses, communication, and other services.

c. **EO**- total requirement for facilities and equipment needed by the Program.

**22. Other Ongoing Projects Being Handled By the Project Leader**- list of ongoing projects being handled by the Project Leader funded by the DOST-GIA Program and other sources, and the accompanying responsibilities relevant to the project.

**23. Other supporting documents required**- as stated in Section VII of DOST A.O. No. 011, Series of 2020 – Revised Guidelines for the Grants-in-Aid Program:

a. Detailed breakdown of the required fund assistance to indicate the counterpart of the proponent and other fund sources including letter/s of commitment from the implementing, collaborating and coordinating agency/entity/ies;[1]
b. A counterpart fund, in kind and/or in cash, shall be required from the implementing agency/entity as one of the application requirements. All projects must have a minimum of 15% counterpart contribution except for projects involving public good;[1]
c. Curriculum Vitae or Personal Data Sheet (PDS) of Project Leader and other co-researchers/implementers. The service record may be requested if needed;[1]
d. Clearance from the DOST or the Funding Agency (e.g., DOST Councils) on previously funded completed projects handled by the Project Leader;[1]
e. Approval from the institution's ethics review board for research involving human subjects or in the case of animal subjects, approval from the Bureau of Animal Industry (BAI) (for PCAARRD- and PCHRD-monitored projects);
f. Clearance from the DOST Biosafety Committee (DOST-BC) shall be required for research proposals involving the use of GMOs under contained use (i.e., experiments done in laboratories, screen house, green house). For projects other than contained use, they shall be referred to the appropriate agency. The DOST Sectoral Councils, after determination as to whether or not the proposal has biosafety implications, shall endorse the same to the DOST-BC in accordance with the prescribed format under Annex 3 of the Philippine Biosafety Guidelines for Contained Use of Genetically Modified Organisms (series of 2014) (if applicable); and
g. For the private non-profit/non-government/people's organizations and startups:
    i. Up-to-date Securities and Exchange Commission (SEC) registration, or Department of Trade and Industry (DTI) registration, or Cooperative Development Authority (CDA) registration certificate, or other authenticated copy of latest Articles of Cooperation and other related legal documents;
    ii. Co-signers Statement (if applicable);
    iii. Copy of latest Income Tax Return;
    iv. Mayor's permit where the business is located;
    v. Audited Financial Statements for the past three (3) years preceding the date of project implementation or in case of those with operation of less than 3 years, for the years in operation and proof of previous implementation of similar projects (or in the case of startups, at least for one (1) year);
    vi. Document showing that NGO/PO has equity to 20 percent of the total project cost, which shall be in the form of labor, land for the project site, facilities, equipment and the like, to be used in the project;
    vii. Disclosure of other related business, if any;
    viii. List and/or photographs of similar projects previously completed, if any, indicating the source of funds for implementation;
    ix. Sworn affidavit of secretary of the NGO/PO that none of its incorporators, organizers, directors or officers is an agent of or related by consanguinity or affinity up to the fourth civil degree to the official of the agency authorized to process and/or approved the proposed MOA, and release of funds;
h. For CSOs, compliance to regulations as required by the General Appropriations Act (GAA) pertaining to fund transfers to Civil Society Organizations (CSOs); and
i. For foundations, DOST certification as accredited by the Science and Technology Foundation Unit

[1] *required of all proposals*

**III. Criteria for Evaluation:**

**A. Criteria for Evaluating Proposals**

| Criterion | Definition |
|---|---|
| Relevance or Significance | Aligned to national S&T priorities, strategic relevance to national development and sensitivity to Philippine political context, culture, tradition and gender and development |
| Technical / Scientific Merit | Sound scientific basis to generate new knowledge or apply existing knowledge in an innovative manner |
| Budget Appropriateness | The proposed budget is commensurate to the proposed work plan and deliverables. |
| Competence of Proponent | Proponent's expertise is relevant to the proposal and with proven competence to implement, manage and complete R&D programs/projects within the approved duration and budget. |

**B. Governing Council / Board and EXECOM's Evaluation Criteria**

| Criteria | Indicators | Raw Score |
|---|---|---|
| A. Soundness of Proposal (20%) | R&D addresses relevant sectoral need (applicable to pressing concern) | 5 |
| | Solution provided is most effective (compared to other proposed solutions) | 5 |
| | Proposed budget is reasonable (project is not expensive vis-a-vis output) | 5 |
| | Work plan is doable in a given timeframe | 5 |
| B. Suitability of Output (30%) | R&D output is cost-effective (cost is competitive in relation to new or existing products or process) | 5 |
| | Has identified partners to adopt the technology (with letter of support from the head of the company) | 5 |
| | Output can be commercialized (through an existing manufacturer, spin-off or start-up company) | 5 |
| | R&D utilization is timely (output should not be overtaken by other solutions) | 5 |
| C. Significance of Outcome (30%) | Economic: increase in productivity, increase in income, new jobs generated, high return of investment (ROI) | 5 |
| | Social: working partnerships established, training opportunities provided, policies adopted, increased access to basic services (i.e., food, health, education); political, cultural, gender sensitivity and inclusivity | 5 |
| | Environment: enhanced environmental health standards, no adverse effect to the environment | 5 |
| | Sustainability: sustainability mechanisms established in terms of institutional, financial and human resources capability (submission of a new proposal to sustain a completed or ongoing proposal does not constitute sustainability of the project) | 5 |
| D. Competence of Proponent (20%) | Proponent's expertise aligned with the proposal | 5 |
| | Collaboration with relevant agencies and/or industry partners | 5 |
| | Thorough understanding of the proposal's deliverables | 5 |
| | DOST has good experience with the proponent | 5 |

C. Additional Criteria on Gender and Development (GAD)